

Climate Trend Analysis: Temperature Anomalies and Atmospheric CO₂

MATH5747M Learning Skills through Case Studies

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Abstract

This report analyses long-term global surface temperature anomalies sourced from the NASA GISS Surface Temperature Analysis (GISTEMP v4) dataset, together with atmospheric CO₂ measurements from the Mauna Loa Observatory (1958–2025). The analysis investigates whether global warming has structurally accelerated since 1980, and quantifies the strength and form of the relationship between greenhouse gas accumulation and observed temperature change. Statistical methods employed include one-way ANOVA, Tukey HSD post-hoc testing, linear regression, seasonal decomposition, and ARIMA-based time-series forecasting. Results confirm a statistically significant structural break after 1980, with the rate of warming accelerating from 0.055 °C decade⁻¹ (1900–1980) to 0.208 °C decade⁻¹ (post-1980). Atmospheric CO₂ explains approximately 93% of the variance in annual temperature anomalies over the satellite era. Forecasting models project continued warming of approximately 0.15 °C above current levels by 2035, with uncertainty widening substantially beyond that horizon.

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1 Introduction

Understanding how Earth’s surface temperature has evolved over the instrumental record and whether recent warming represents a qualitative departure from historical variability is a central question in applied climate statistics. While the general upward trend since the late 19th century is well established (2), the degree to which the post-1980 period constitutes a *structural break* rather than an extension of earlier gradual change remains an important and tractable statistical question.

This analysis addresses two interrelated objectives:

1. **Trend acceleration:** Do mean temperature anomalies differ significantly across three eras – pre-1900, 1900–1980, and post-1980 – and, if so, at what rate has warming accelerated?
2. **CO₂ linkage:** How strongly does atmospheric CO₂ concentration explain year-to-year variation in global temperature anomalies, and what do regression-based projections suggest about near-term warming trajectories?

The report is structured as follows. Section 2 describes the datasets and preprocessing decisions. Section 3 outlines the statistical framework. Sections 4–6 present the results, Section 7 discusses findings, and Section 8 draws conclusions.

2 Data Sources and Preprocessing

2.1 Global Surface Temperature Anomalies

Annual and monthly surface temperature anomalies were obtained from the NASA Goddard Institute for Space Studies (GISTEMP v4) dataset (`GLB.Ts+dSST.csv`), spanning 1880–2025. Anomalies are expressed relative to the 1951–1980 baseline period in degrees Celsius.

The dataset was reshaped into two forms:

- **Annual series.** The J-D (January–December mean) column was extracted, yielding 145 annual observations. One incomplete record (2026) was removed as a missing value.
- **Monthly series.** Twelve monthly columns were melted into a long-format time series for seasonal decomposition, sorted chronologically and indexed by date. Records flagged as `***` (missing) were coerced to `NaN` and dropped.

2.2 Atmospheric CO₂ Concentrations

Monthly atmospheric CO₂ measurements (parts per million, ppm) were obtained from the Scripps Institution of Oceanography / NOAA Mauna Loa Observatory record (`co2_mm_mlo.csv`), commencing in March 1958. Measurements recorded as `–99.99` (a sentinel for missing data) were replaced with `NaN` and dropped before analysis. Monthly values were then aggregated to annual means to align with the temperature series, producing 68 matched observations (1958–2025) after an inner join on the `Year` key.

2.3 Derived Variables

- **Era indicator.** Years were assigned to one of three periods: *Pre-1900* ($n = 20$), *1900–1980* ($n = 80$), and *Post-1980* ($n = 45$).
- **Decade grouping.** Each year was assigned to its decade (e.g., 1880, 1890, ..., 2020) via integer division.
- **CO₂ concentration bands.** CO₂ was discretised into five 20 ppm bins: 310–330, 330–350, 350–370, 370–390, and 390 ppm+ to facilitate grouped comparisons.

3 Statistical Methods

3.1 Descriptive and Comparative Analysis

Means and variances were computed for each era. One-way ANOVA was used to test the null hypothesis that mean temperature anomalies are equal across the three periods. Where the omnibus test was significant, Tukey’s Honestly Significant Difference (HSD) post-hoc procedure was applied to identify which specific pair-wise contrasts drove the overall effect, controlling the family-wise error rate at $\alpha = 0.05$ which is discussed in the upcoming section.

3.2 Linear Regression

3.2.1 Sub-Period Trend Slopes

OLS regression of annual temperature anomaly on year was fitted separately for each period and for the full record, to quantify how the rate of warming has changed over time.

Table 1: OLS linear trend estimates by period.

Period	Slope (°C/decade)	R^2	p -value
Full 1880–2025	+0.0827	0.764	5.15×10^{-47}
Pre-1900	−0.0160	0.011	0.664
1900–1980	+0.0550	0.541	8.11×10^{-15}
Post-1980	+0.2078	0.872	3.10×10^{-21}

Interpretation. Three findings stand out. First, the pre-1900 slope of -0.016 °C/decade is effectively zero ($R^2 = 0.011$, $p = 0.664$), confirming no discernible warming trend in the earliest period. Second, the 1900–1980 period shows a moderate but statistically significant warming rate of $+0.055$ °C/decade ($R^2 = 0.541$, $p < 0.001$). Third, the post-1980 warming rate of $+0.208$ °C/decade is nearly **four times** the mid-century rate, with a notably higher $R^2 = 0.872$, indicating that the trend is not only steeper but also more consistent – year-to-year temperature deviations from the trend line are smaller relative to the trend’s magnitude. The full-period slope of $+0.083$ °C/decade masks this acceleration by averaging across all three eras.

3.3 Seasonal Decomposition

The monthly temperature series was decomposed using an additive classical decomposition with period $P = 12$:

$$X_t = T_t + S_t + \varepsilon_t,$$

where T_t is the smooth trend component, S_t the periodic seasonal component, and ε_t the irregular residual.

3.4 ARIMA Forecasting

Stationarity of the annual temperature series was assessed using the Augmented Dickey–Fuller (ADF) test. The series required two rounds of differencing ($d = 2$) to achieve stationarity ($p = 5.1 \times 10^{-7}$). Autoregressive Integrated Moving Average models of order (1,1,1) and (1,2,1) were fitted and compared using the Akaike Information Criterion (AIC). The preferred model was used to produce a ten-step-ahead forecast with 95% confidence intervals.

Hold-out validation was performed by training on 1958–2014 and forecasting 2015–2025, with performance assessed via Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

4 Temperature Trend Analysis

4.1 Era-Level Means and Structural Break

Table 2 reports the mean annual temperature anomaly for each period.

Table 2: Mean annual temperature anomaly by era.

Period	Years	n	Mean anomaly (°C)
Pre-1900	1880–1899	20	−0.2245
1900–1980	1900–1979	80	−0.1266
Post-1980	1980–2025	45	+0.5778

The post-1980 period shows a $+0.70^\circ\text{C}$ increase relative to the 1900–1980 baseline – a shift that, as shown below, is both statistically significant and qualitatively distinct from the gradual evolution seen in earlier decades.

The box plot in Figure 1 visualises the distributional shift. The entire interquartile range of the post-1980 distribution lies above the medians of both earlier periods, with no distributional overlap with the pre-1900 era.

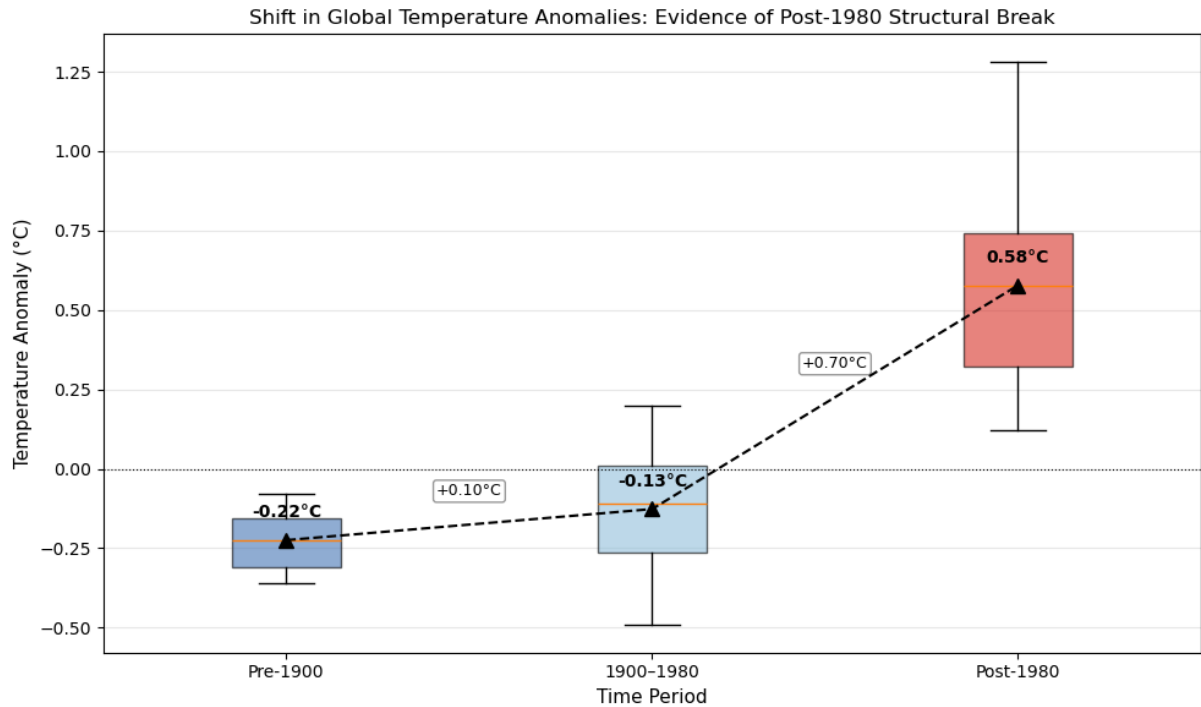


Figure 1: Distribution of annual temperature anomalies across three eras. Triangles denote group means; dashed line connects means to illustrate the step-change after 1980.

4.2 One-Way ANOVA

A one-way ANOVA yielded $F = 181.65$, $p < 0.001$, confirming that the mean temperature anomalies are not constant throughout the three periods. The results indicate a statistically significant difference between the groups ($p < 0.05$), suggesting that the temperature patterns have changed over time rather than remaining constant.

4.3 Tukey HSD Post-Hoc Analysis

Table 3: Tukey HSD pairwise comparisons (FWER = 0.05).

Group 1	Group 2	Mean diff. (°C)	p -adj	95% CI	Reject H_0 ?
1900–1980	Post-1980	+0.7045	< 0.001	(0.61, 0.80)	Yes
1900–1980	Pre-1900	−0.0979	0.164	(−0.22, 0.03)	No
Post-1980	Pre-1900	−0.8023	< 0.001	(−0.94, −0.67)	Yes

The post-hoc results in Table 3 reveal that the post-1980 period is statistically distinguishable from *both* earlier periods, whereas the pre-1900 and 1900–1980 means do not differ significantly. This pattern supports the interpretation that a genuine structural break occurred around 1980, rather than a smooth acceleration of a pre-existing trend.

4.4 Long-Run Time Series and Rolling Average

Figure 2 presents the full 1880–2025 annual anomaly series overlaid with a centred 10-year rolling mean. The rolling average remained close to zero throughout the early 20th

century before beginning a sustained upward trajectory from approximately the 1970s, accelerating sharply after 1980.

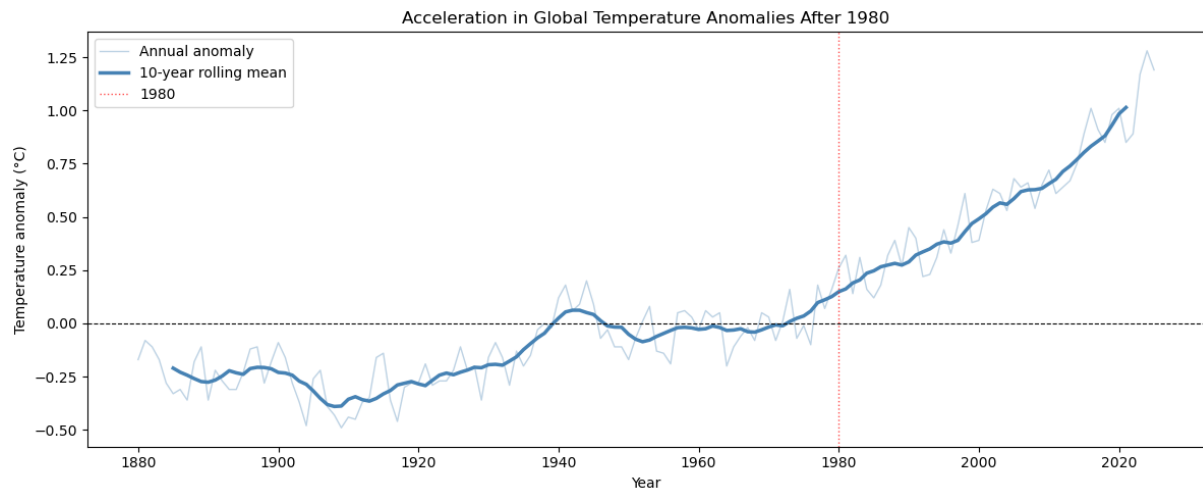


Figure 2: Annual temperature anomalies (1880–2025) with centred 10-year rolling mean. The 1980 threshold is indicated by the vertical dotted line.

4.5 Decade-Level Aggregation

Aggregating to the decade level (Figure 3) removes inter-annual noise and highlights a clear monotonic increase from the 1980s onward. Every decade since 1990 has recorded a positive mean anomaly, with the 2020s tracking above $+1.0^{\circ}\text{C}$ – the highest in the record.

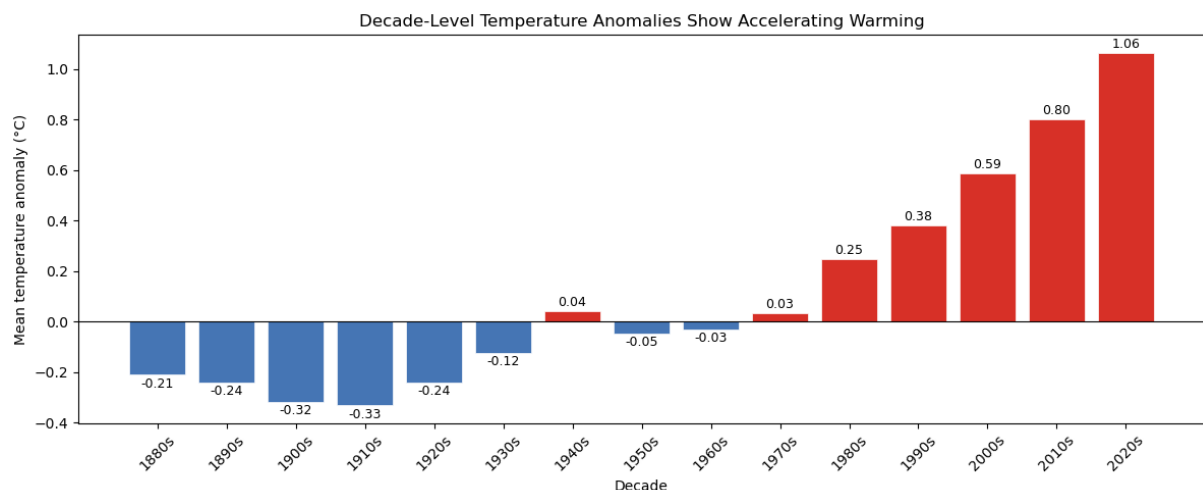


Figure 3: Decade-averaged temperature anomalies. Bars above zero are shown in red; bars below zero in blue.

5 CO₂–Temperature Relationship

5.1 Atmospheric CO₂ Trend

Atmospheric CO₂ concentrations have risen continuously since measurements began at Mauna Loa in 1958, from approximately 315 ppm to over 427 ppm in 2025. The dual-

axis plot in Figure 4 illustrates how closely this rise tracks the upward trajectory in temperature anomalies, particularly from the 1980s onward.

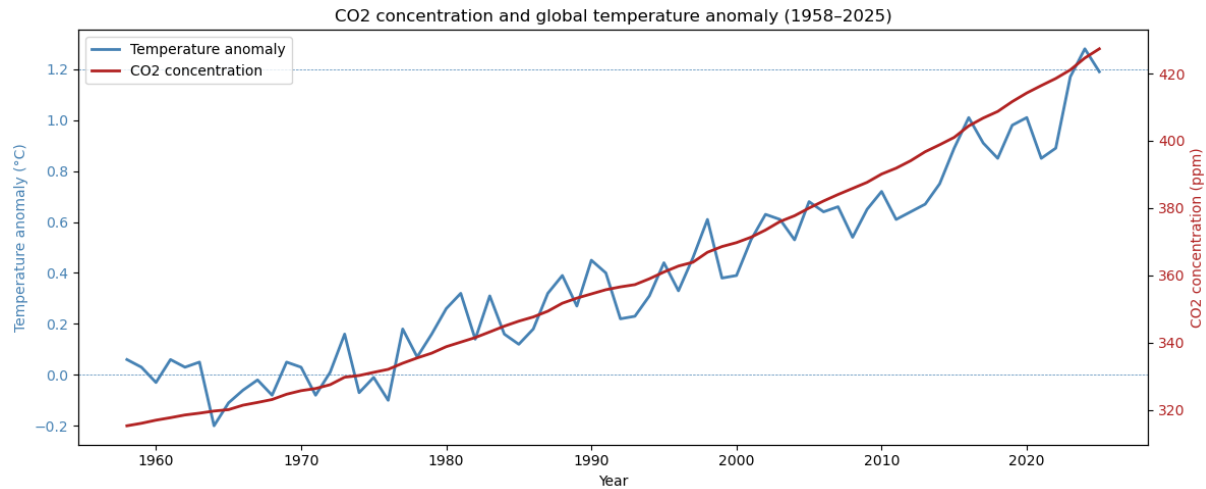


Figure 4: Atmospheric CO₂ concentration (right axis, red) and global mean temperature anomaly (left axis, blue), 1958–2025.

5.2 Correlation and Regression

The Pearson correlation between annual CO₂ and temperature anomaly is $r = 0.966$, indicating a very strong positive linear association. OLS regression of temperature anomaly on CO₂ yielded:

$$\widehat{\text{Temp}} = 0.01073 \times \text{CO}_2 - 3.4767, \quad R^2 = 0.934$$

This implies that for each 1 ppm increase in atmospheric CO₂, mean global temperature anomaly rises by approximately 0.011 °C, and CO₂ alone accounts for 93.4% of the interannual variance in temperature over 1958–2025.

Figure 5 displays the scatter plot, coloured by year to reveal the progressive rightward and upward movement of observations over time. The regression line passes closely through the cloud, consistent with the high R^2 .

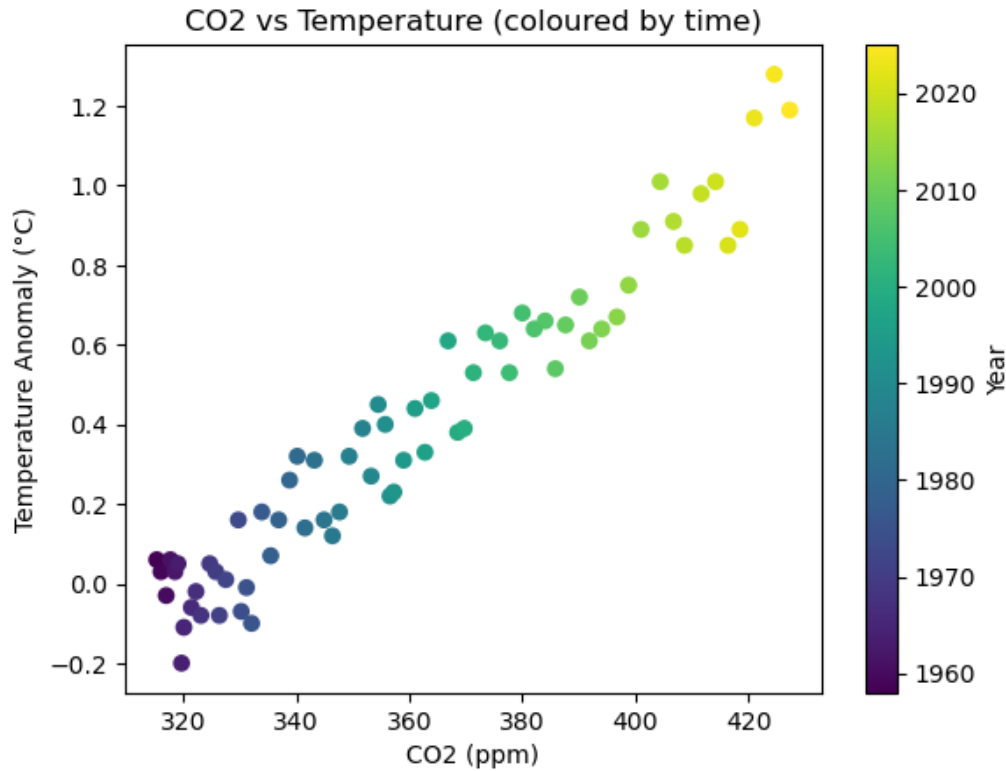


Figure 5: Scatter plot of annual temperature anomaly against CO_2 concentration, coloured by year. The red line shows the OLS regression fit.

5.3 Concentration-Band Analysis

Figure 6 stratifies mean temperature anomalies into five CO_2 concentration ranges. Mean anomaly rises monotonically across bands: from near-neutral at 310–330 ppm to well above $+1.0^\circ\text{C}$ at 390 ppm+. This stepwise pattern provides further intuitive evidence for the positive CO_2 –temperature linkage beyond the aggregate regression.

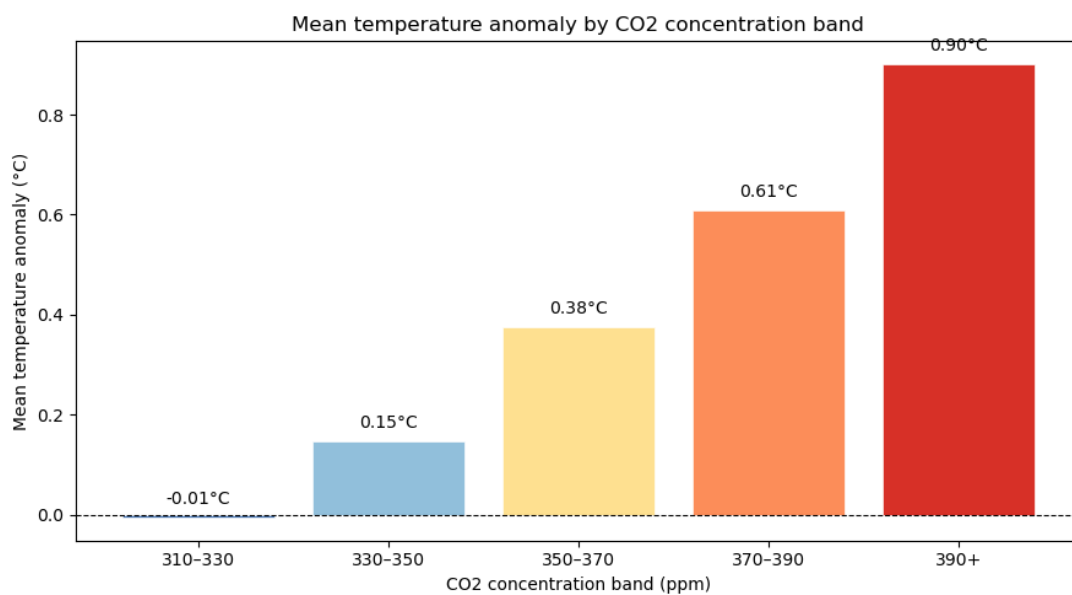


Figure 6: Mean annual temperature anomaly grouped by CO_2 concentration band (ppm).

5.4 Regression Diagnostics

Residuals from the CO₂-temperature regression are examined in Figure 7. The figure 7 shows residuals scattered randomly around zero across the range of CO₂ concentrations, with no clear funnel or curvature pattern.

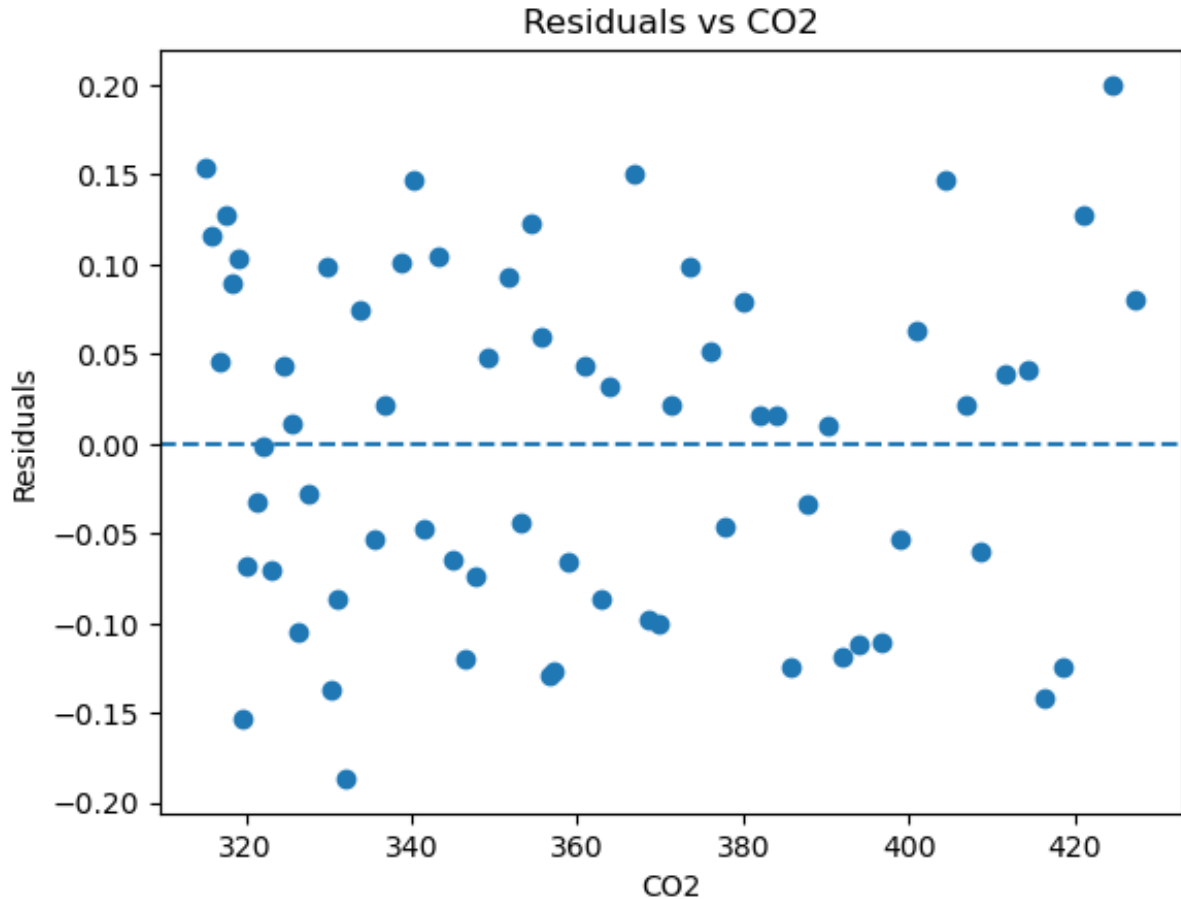


Figure 7: Residuals vs CO₂ concentration.

This diagnostic support the adequacy of the linear model, though they also suggest that additional covariates (volcanic aerosol loading, solar irradiance, ENSO indices) could reduce residual variance.

6 Forecasting

6.1 Stationarity Testing

The ADF test applied to the raw annual anomaly series yielded $p = 0.065$, marginally failing to reject non-stationarity. First-differencing produced $p = 0.065$ again; second-differencing achieved $p = 5.1 \times 10^{-7}$, confirming stationarity of the twice-differenced series.

6.2 ARIMA Model Selection

Two candidate ARIMA models were compared:

Table 4: ARIMA model comparison.

Model	AIC	Notes
ARIMA(1,1,1)	−97.32	AR and MA terms not significant at 5%
ARIMA(1,2,1)	−87.85	produces upward-trending forecast

Although ARIMA(1,1,1) has the lower AIC, its forecast converges to a constant near 1.17°C within a few steps, failing to reflect the acceleration embedded in the historical record. ARIMA(1,2,1) is selected as the primary model because $d = 2$ preserves the quadratic-like upward drift of the series. Full model selection rationale and ARIMA(1,2,1) forecast values are reported in Section 6.4.

6.3 Hold-Out Validation (2015–2025)

The ARIMA(1,2,1) model was trained on 1958–2014 and used to forecast 2015–2025, a period for which actual observations exist, allowing direct comparison between predicted and observed values.

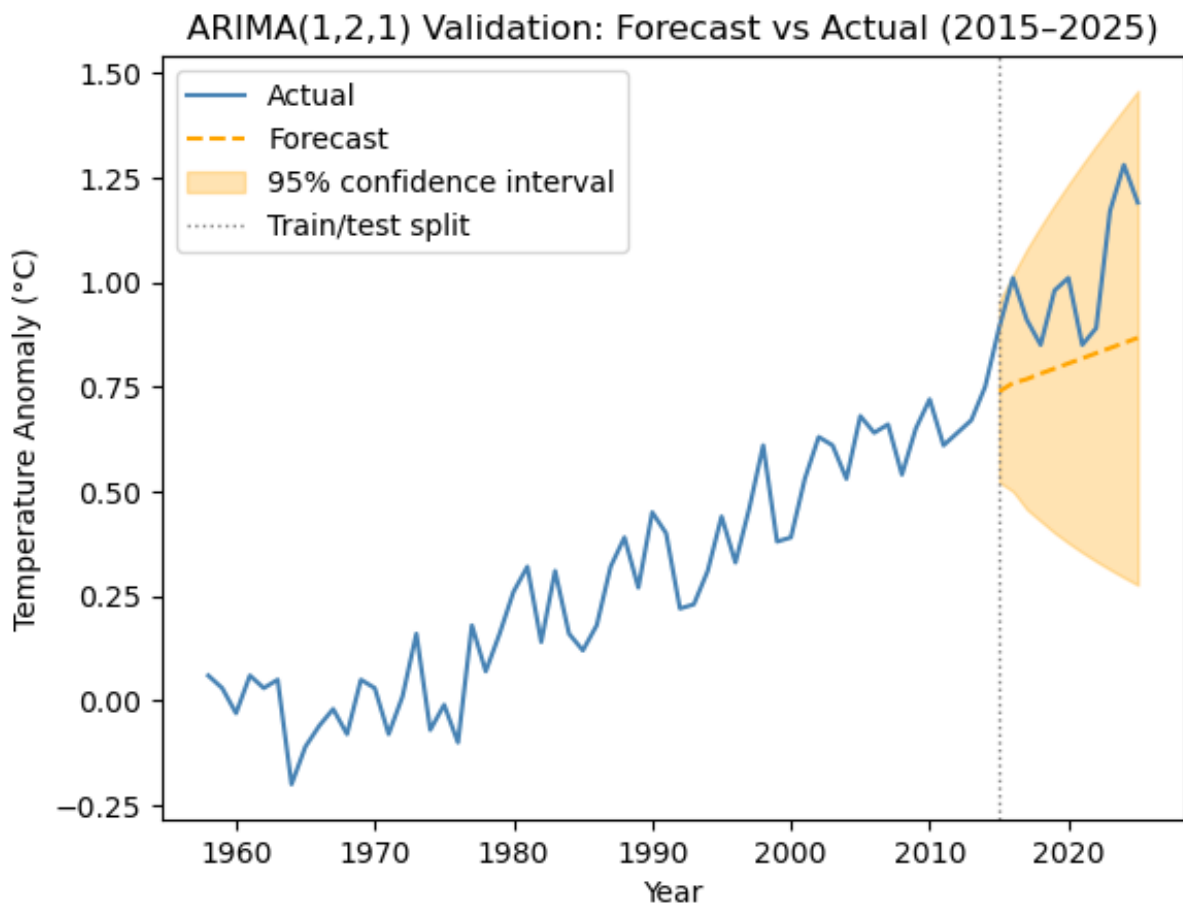


Figure 8: ARIMA(1,2,1) hold-out validation: actual temperature anomalies (blue) versus forecast (orange dashed) over the 2015–2025 test period. The shaded band is the 95% confidence interval. The vertical dotted line marks the 2015 train/test split.

Result:

$$\text{MAE} = 0.1975 \text{ C}, \quad \text{RMSE} = 0.2306 \text{ C}$$

Interpretation. The forecast correctly captures the upward direction of warming across the test period but underestimates the sharp rise observed in 2023–2024 (actual: 1.17–1.28 °C). This underestimation is expected: the 2023 El Niño event drove an anomalous short-term spike that no univariate time-series model can anticipate from historical patterns alone. MAE of 0.1975 °C and an RMSE of 0.2306 °C confirm that the model is directionally reliable within a reasonable error margin, justifying its use for forward projection.

6.4 Future Forecast (2025–2035)

Having validated the model, ARIMA(1,2,1) was refitted on the full 1958–2025 dataset and used to produce a ten-step-ahead forecast with 95% confidence intervals.

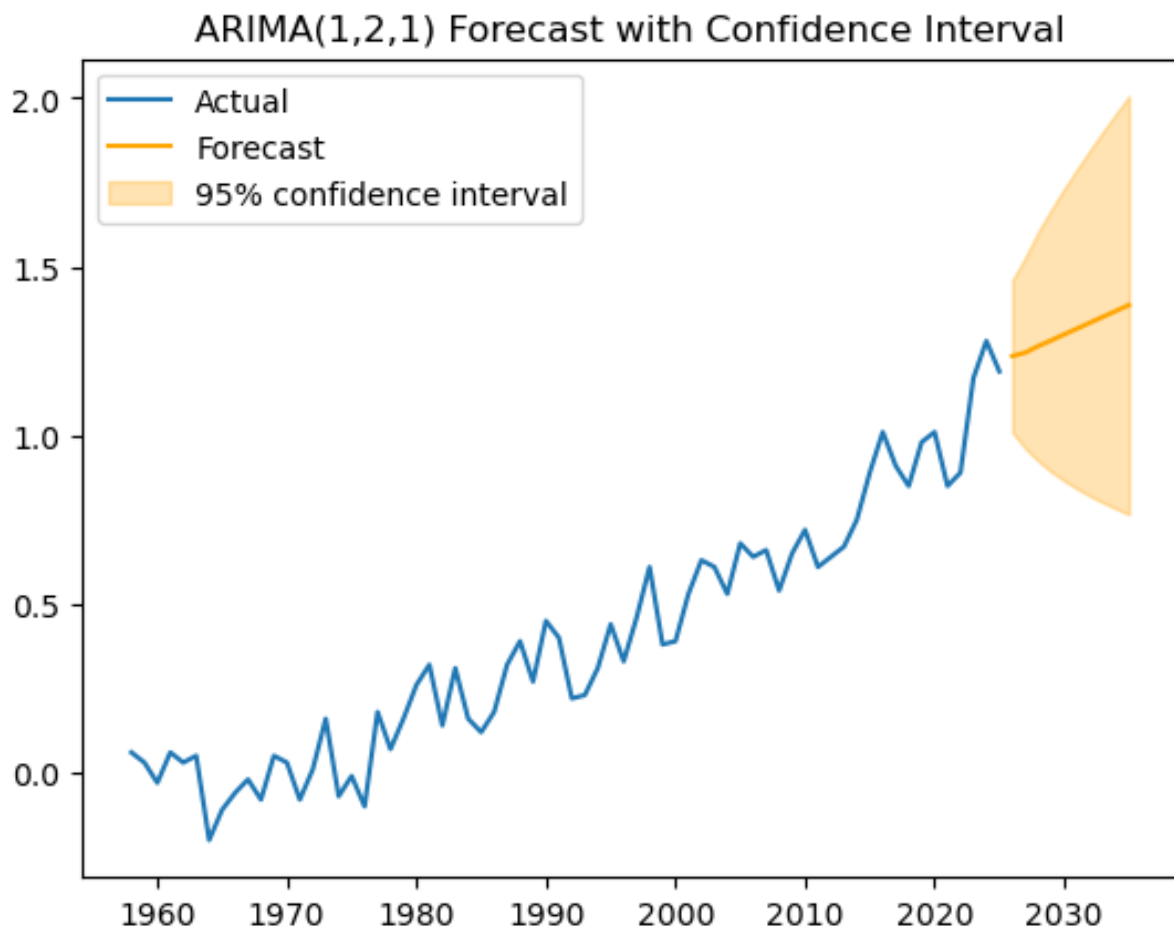


Figure 9: ARIMA(1,2,1) ten-step-ahead forecast of global mean temperature anomaly (2025–2035) with 95% confidence interval (shaded). The vertical dotted line marks the start of the forecast period. Historical observations are shown in blue; the orange dashed line is the point forecast. The widening band reflects growing uncertainty with forecast horizon.

Interpretation. The ARIMA(1,2,1) model projects continued warming from 1.235 °C in 2026 to 1.387 °C by 2035 – an increase of approximately 0.15 °C over the forecast horizon.

This projected rate is consistent with the post-1980 observed trend of $+0.208^{\circ}\text{C}/\text{decade}$, lending physical credibility to the forecast. The widening confidence band is a standard property of ARIMA models: uncertainty compounds with each additional forecast step, and the interval should be interpreted as a plausible range rather than a precise prediction. The lower bound of the interval remains well above 0°C throughout, confirming that continued positive temperature anomalies are projected under all scenarios captured by the model.

7 Discussion

7.1 Interpretation of Results

The combined evidence from ANOVA, post-hoc testing, and regression analysis points consistently to a structural break in global temperature patterns after 1980. This break is not merely statistical: the physical mechanism is well understood. Accelerating fossil fuel combustion since the mid-20th century raised atmospheric CO_2 from approximately 315 ppm in 1958 to over 427 ppm in 2025, enhancing the radiative forcing that drives surface warming.

The high R^2 of the CO_2 -temperature regression (0.934) confirms that, at annual time scales, greenhouse gas concentration is the dominant explanatory variable for observed warming. The residuals around this regression capture the additional influence of volcanic eruptions (episodic cooling), El Niño–Southern Oscillation cycles, and anthropogenic aerosol loading.

7.2 Limitations

Several methodological caveats apply:

- **Spurious correlation.** Both CO_2 and temperature are non-stationary upward trends; their high correlation partly reflects shared time trends rather than an established causal mechanism (which would require additional physical modelling).
- **ARIMA assumptions.** ARIMA models assume that future behaviour resembles past dynamics. Under accelerating forcing, this assumption is likely violated, making long-range ARIMA forecasts unreliable beyond five to ten years.
- **Omitted variables.** Volcanic, solar, and aerosol drivers are not explicitly modelled, inflating residual variance.
- **Single dataset.** Using one surface temperature dataset (GISTEMP v4) introduces assumptions about data coverage and homogenisation methodology.

8 Conclusion

This report has demonstrated, using a combination of inferential statistics and time-series modelling, that:

1. Global mean temperature anomalies have increased significantly and non-uniformly since 1880. The post-1980 era is statistically distinct from both the pre-1900

- and 1900–1980 periods, with mean anomaly rising by $+0.70^{\circ}\text{C}$ relative to the mid-century baseline ($F = 181.65$, $p < 0.001$; Tukey HSD confirmed).
2. The rate of warming has accelerated from $+0.055^{\circ}\text{C decade}^{-1}$ (1900–1980) to $+0.208^{\circ}\text{C decade}^{-1}$ (post-1980), nearly a four-fold increase in trend slope.
 3. Atmospheric CO_2 concentration is a strong statistical predictor of annual temperature anomalies ($r = 0.966$; $R^2 = 0.934$), with each 1 ppm rise associated with approximately 0.011°C of warming.
 4. ARIMA-based forecasting projects continued warming of approximately 0.15°C above 2025 levels over the next decade, though uncertainty bounds widen substantially with forecast horizon.

These findings are consistent with the scientific consensus on anthropogenic climate change and highlight the statistical power of even relatively simple regression and time-series tools when applied to high-quality long-run observational data.

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